

Predicting Europe's Daily Weather Patterns

Group 15

Introduction

This project examines whether historical weather observations can accurately predict next-day mean temperatures across 18 European locations. The dataset contains 3,654 daily observations collected between January 2000 and January 2010 and includes measurements such as temperature, humidity, pressure, wind speed, and wind gust. To improve prediction accuracy, new features including lagged variables, rolling averages, interaction terms, and calendar-based variables were engineered from the original data. Decision Tree and Random Forest regression models were trained using a chronological train-test split and compared with a mean baseline model. Model performance was evaluated using mean absolute error (MAE), mean squared error (MSE), coefficient of determination (R^2), and time-series cross-validation. The Random Forest produced the most accurate predictions, achieving a test MAE of 1.58°C and an R^2 of 0.92.

Data Source and Preparation

The Weather Prediction dataset was obtained from Kaggle and contains daily weather observations from 18 European cities and was designed for machine learning applications. It includes 165 variables describing temperature, humidity, pressure, wind speed, wind gust, precipitation, sunshine, and other atmospheric measurements. Not every variable was available for every city, requiring the analysis to dynamically identify usable predictors.

A collection of lagged, rolling, interaction, and difference features were generated in order for the model to have usable data and understand how they can be combined to predict future outcomes. Rows affected by lagged and rolling features were removed while missing predictor values were handled using median imputation. We chronologically split the data into two, with training data spanning 2000 through 2008 and testing data being from 2008 to 2010.

Models

Three models were evaluated: Random Forest, a Decision Tree regressor, and a mean baseline. The Decision Tree used a maximum depth of 10 and a minimum split size of 4. TimeSeriesSplit cross-validation was used to estimate model performance during training while preserving the chronological ordering of the observations.

The Random Forest model used an ensemble of 300 Decision Tree regressors, with each tree set to a maximum depth of 20 and with at least 4 samples at any node before performing a split. These constraints reduce overfitting while still allowing the model to capture complex nonlinear relationships in the meteorological features.

Results

The Random Forest produced the best overall performance. It achieved a test MAE of 1.58°C , meaning predicted temperatures differed from observed temperatures by about 1.6°C on average. The model also obtained a test MSE of 4.11 and an R^2 of 0.92, indicating that it explained approximately 92% of the variation in daily mean temperatures across the test data. Cross-validation produced a mean MSE of 5.05, suggesting that the model generalized well and showed limited overfitting.

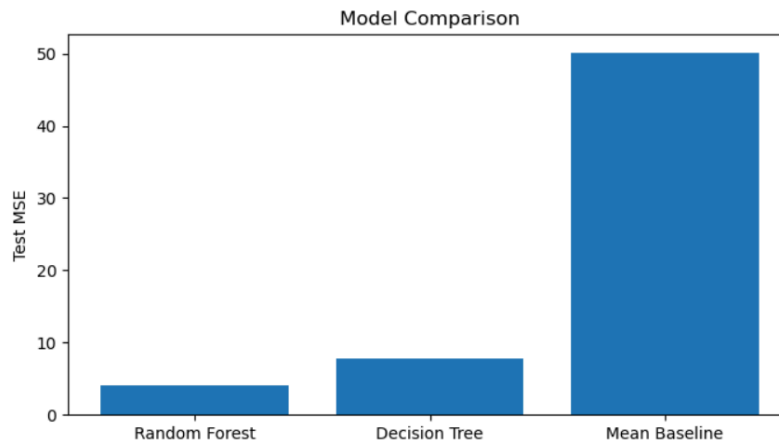
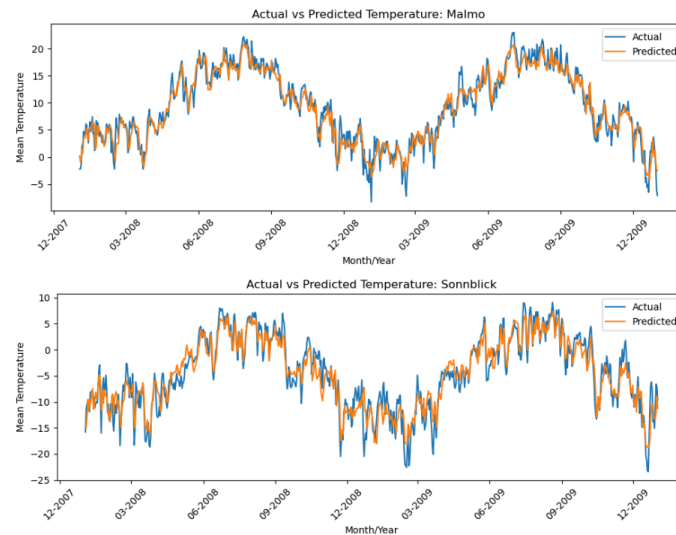


Figure 1. Test MSE for the Random Forest, Decision Tree, and mean baseline



Conclusion

Historical weather observations can accurately predict next-day mean temperatures across multiple European cities when combined with engineered temporal features and Random Forest regression. The model substantially outperformed both the Decision Tree and the baseline model, indicating that the ensemble approach effectively captured nonlinear relationships within the data. Future work could evaluate additional forecasting models, perform systematic hyperparameter optimization using `RandomizedSearchCV`, and incorporate additional meteorological variables to improve prediction accuracy.

Contributions

Member	Proposal	Coding	Presentation	Report
Santos Mejia Ayala	0	0.6	1	1
Giang Tran	0	1	1	0.5
Ana-Lucette Naranjo-Fernandez	1	1	1	0.5
Hera Salehi	1	0	0.5	0
Benjamin Verhaalen	0	0	0.5	0

Note:

- In the chart above, 1 = full contribution, 0.1 - 0.9 = partial, 0 = no contribution